EVERY ALGEBRA OVER A FIELD CONTAINS A MAXIMAL WEAKLY ETALE SUBALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The author writes '(previous lemma cited)' in place of an explicit label at proof line 30907; that wording is delivered verbatim and no label was supplied in its place, though the sentence is still recorded in external_references as a navigation pointer. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The class of weakly etale algebras over a field is closed under tensor products and quotients. invoked at proof lines 30899-30901. Supplies the directedness of the collection of weakly etale subalgebras.
- The characterization of weakly etale algebras, in particular their stability under filtered colimits. invoked at proof lines 30904-30905. Shows the colimit of the directed system is again weakly etale.
- The preceding statement on the image of a weakly etale algebra map. invoked at proof lines 30906-30907. Shows the image of B_max in A is weakly etale, giving the maximal element.

Source reference. Original lemma 30846: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30846–30846. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30454: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30454–30454. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let K be a field. Let A be a K-algebra. There exists a maximal weakly étale K-subalgebra $B_{max} \subset A$.*

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Proof. Let $B_1, B_2 \subset A$ be weakly étale K-subalgebras. Then $B_1 \otimes_K B_2$ is weakly étale over K and so is the image of $B_1 \otimes_K B_2 \to A$ (Lemma 30846). Thus the collection $\mathcal{B}$ of weakly étale K-subalgebras $B \subset A$ is directed and the colimit $B_{max} = \operatorname{colim}_{B \in \mathcal{B}} B$ is a weakly étale K-algebra by Lemma 30454. Hence the image of $B_{max} \to A$ is weakly étale over K (previous lemma cited). It follows that this image is in $\mathcal{B}$ and hence $\mathcal{B}$ has a maximal element (and the image is the same as B_{max}). $\square$